

INSTRUCTIONAL DESIGN PLAN (IDP)

Six Sigma Master Black Belt (MBB) Certificate Program

1. Document Control Information

Program Name	Six Sigma Master Black Belt (MBB) Certificate Program
Certificate Type	Professional Certificate Program
Program Category	Quality Management / Process Improvement / Lean Six Sigma
Document Title	Instructional Design Plan (IDP)
Organization	International Management Consultancy L.L.C. (IMC Certifications)
Prepared By	Bhuvaneshwari Premkumar
Reviewed By	Mr.Farhan Ahmad
Approved By	Mr.Farhan Ahmad
Date Created	2/6/2026
Date Updated	4/13/2026
Distribution	Internal / Accreditation Evidence

2. Purpose of the Instructional Design Plan

The purpose of this Instructional Design Plan (IDP) is to describe the structured instructional design methodology used in developing the Six Sigma Master Black Belt (MBB) Certificate Program.

This document serves as evidence that the program has been developed using a systematic process aligned with internationally recognized best practices and is intended to support compliance with ASTM E2659-18 requirements for certificate programs.

This IDP outlines the program's design, learning outcomes, content development approach, assessment blueprint, examination development methodology, implementation process, and continuous improvement mechanisms.

3. Program Overview

3.1 Program Description

The Six Sigma Master Black Belt (MBB) Certificate Program is an advanced-level professional certification designed for experienced Six Sigma practitioners who aim to lead enterprise-level process improvement initiatives and mentor Lean Six Sigma teams.

The program emphasizes advanced statistical analysis, project leadership, strategic deployment of Six Sigma, and governance frameworks required to manage organization-wide continuous improvement programs.

This program provides learners with both theoretical knowledge and practical competency required to function as Master Black Belts within large organizations.

3.2 Program Aim

The aim of this program is to develop highly competent professionals who can lead complex improvement programs, guide Black Belt projects, drive operational excellence strategy, and support business transformation initiatives.

3.3 Program Objectives

Upon completion of this program, learners will be able to:

- Develop and manage Six Sigma deployment strategies aligned with organizational objectives.
- Lead and mentor multiple Black Belt and Green Belt project teams.
- Apply advanced statistical tools and methodologies to solve complex business problems.
- Evaluate and optimize organizational processes using Lean and Six Sigma frameworks.
- Design governance and sustainability models for continuous improvement programs.

4. Target Audience

The Six Sigma Master Black Belt program is designed for:

- Certified Six Sigma Black Belts with significant project experience
- Quality Managers / Operational Excellence Leaders
- Process Improvement Managers
- Business Excellence Professionals
- Manufacturing, Healthcare, Service Industry Quality Professionals
- Continuous Improvement Consultants

Recommended Experience Level

Candidates are expected to have:

- Prior completion of Lean Six Sigma Black Belt certification
- Five years of professional work experience
- Maintain a minimum 80% attendance in all program sessions.
- Achieve at least 80% on summative assessments, aligned with the program's intended learning outcomes.

5. Program Scope

The scope of this program includes advanced Six Sigma concepts, leadership capabilities, statistical applications, project governance, mentoring methodology, deployment planning, and strategic improvement initiatives.

The program includes:

- Advanced DMAIC tools and project execution methodology
- Advanced statistical analysis techniques
- Lean enterprise and waste elimination frameworks
- Change management and organizational deployment strategies
- Coaching and mentoring methodologies
- Project evaluation and sustainability planning

6. Certificate Term / Validity

The Six Sigma Master Black Belt Certificate issued under this program is valid lifetime

7. Requisites for Earning the Certificate

To successfully earn the Six Sigma Master Black Belt certificate, candidates must meet the following requirements:

- Three years certified Lean Six Sigma Black Belt
- Five years of professional work experience
- Accomplished five professional development activities
- Completed three projects or kaizen events detailing
- Maintain a minimum 80% attendance in all program sessions.
- Achieve at least 80% on summative assessments, aligned with the program's intended learning outcomes.
- Optional: Writing and examiner duties can be submitted to strengthen application if desired.

8. Instructional Design and Development Team

8.1 Individuals Involved in Developing the Program

The program was developed collaboratively by an instructional design and subject matter expert (SME) team consisting of quality management professionals, certified Lean Six Sigma experts, instructional designers, and program reviewers.

Name	Role	Responsibility
Shashi Prakash	Lead Instructional Designer	Learning design, mapping, assessment alignment
Shashi Prakash	SME / Master Black Belt	Content development, review, technical accuracy
Deepa Meghwani	Program Reviewer	Validation of structure and compliance

Bhuvaneshwari	Assessment Developer	Exam blueprint and question development
Mr.Farhan Ahmad	Quality Assurance Reviewer	Continuous improvement alignment

8.2 Qualifications of Individuals

All individuals involved in program development possess relevant education, professional certifications, and experience aligned with Lean Six Sigma training and process improvement domains.

Lead SME : Shashi Prakash

- Education : Certified MBB
- Certifications: Master Black Belt / PMI
- Experience: 20 years in Six Sigma deployment
- Training Experience: 10 years as trainer/consultant

Quality Assurance Reviewer : Farhan Ahmad

- Education : Certified MBB
- Certifications: Master Black Belt
- Experience: 5 years in Six Sigma deployment
- Training Experience: 5 years consultant

Program Reviewer: Deepa Meghwani

- Education: Bachelor of Design
- Certifications: Master Black Belt / PMI
- Experience: 15 Years of Program Compliance
- Training Experience: 5 years consultant

Assessment Developer : Bhuvaneshwari Premkumar

- Education : Bachelor of Computer Science
- Certifications: ISTD Certified L&D , Certified Project Manager, CIPD Candidate
- Experience: 11 Years Training and Development & Learning Design
- Training Experience : 10 Years or Product and Process Training

9. Needs Analysis Process

9.1 Needs Analysis Methodology

The need for the Six Sigma Master Black Belt program was identified through a structured needs analysis process, including:

- Market research and benchmarking of global Six Sigma advanced-level certifications
- Review of industry requirements for operational excellence leadership roles
- Internal corporate client requirements and training inquiries
- Feedback from previous Six Sigma learners and corporate stakeholders
- Consultation with subject matter experts and senior quality professionals
- Review of job market trends and role expectations (e.g., Operational Excellence Manager, CI Director)

9.2 Summary of Needs Analysis Findings

The needs analysis identified the following gaps and market demands:

- Organizations require leaders capable of driving enterprise-level Lean Six Sigma deployments.
- Many Black Belts possess technical skills but lack mentoring and governance capabilities.
- Demand is increasing for professionals who can integrate Six Sigma with strategy and change management.
- Organizations require strong capability in advanced statistics, hypothesis testing, regression, and multivariate analysis.
- Sustainability of Six Sigma programs requires leadership in project selection, prioritization, and benefits realization.

10. Development of Intended Learning Outcomes (ILOs)

The Intended Learning Outcomes (ILOs) for this program were developed based on:

- Results of the needs analysis
- Industry competency frameworks
- Benchmarking with international Six Sigma bodies and best practices
- Bloom's Taxonomy (Cognitive Levels: Application, Analysis, Evaluation, Creation)
- Alignment with real-world project requirements and organizational deployment needs

The ILOs were finalized through review and validation by SMEs and program reviewers.

11. Intended Learning Outcomes (ILOs)

Upon successful completion of this program, learners will be able to:

1. Evaluate organizational readiness for Six Sigma deployment and recommend improvement strategies.
2. Design Six Sigma governance structures including project selection, tracking, and benefits realization.
3. Apply advanced statistical tools such as regression, ANOVA, hypothesis testing, and DOE for process improvement.
4. Lead complex DMAIC projects involving cross-functional stakeholders and high-impact business problems.
5. Develop mentoring frameworks to coach Black Belts and Green Belts in project execution.

6. Assess Lean integration opportunities and optimize value streams using Lean enterprise principles.
7. Create sustainability mechanisms to ensure continuous improvement initiatives remain effective long-term.
8. Critically evaluate process improvement outcomes and validate results using statistical evidence.

12. Link between ILOs and Course Modules (Curriculum Mapping)

The following curriculum mapping demonstrates the alignment between the Intended Learning Outcomes (ILOs) and the course modules. The mapping ensures that each ILO is addressed through appropriate learning content, tools, and applied learning components. This alignment is supported by the needs analysis findings and evidence gathered from job postings, competitor benchmarking, and competency framework reviews.

ILO	ILO Statement (Summary)	Primary Module	Supporting Module(s)
ILO 1	Evaluate readiness for Six Sigma deployment	Module 1	Module 6
ILO 2	Design governance structures (project selection, tracking, benefits)	Module 1	Module 6
ILO 3	Apply advanced statistical tools (Regression, ANOVA, DOE)	Module 3	Module 7
ILO 4	Lead complex DMAIC projects with stakeholders	Module 2	Module 5, Module 7
ILO 5	Develop mentoring frameworks to coach BB/GB	Module 5	Module 2
ILO 6	Assess Lean integration and optimize value streams	Module 4	Module 2
ILO 7	Create sustainability mechanisms for CI long-term	Module 6	Module 1
ILO 8	Evaluate outcomes and validate statistically	Module 3	Module 7

13. Course Content Development

13.1 Content Development Methodology

Course content was developed through a structured process that ensured accuracy, relevance, and alignment with intended learning outcomes.

The content development process included:

- SME preparation of technical content and topic structure
- Instructional design team conversion of content into learner-friendly format (PPT edits)
- Development of case studies, examples, and activities
- Internal review and proofreading
- Alignment checks with ILOs and assessment blueprint

13.2 Content Developers and Qualifications

Name	Role	Qualification	Experience
Shashi Prakash	SME	Trainer	20
CPD	Approving body	Organization	NA
Bhuvaneshwari	L&D	Diploma In L&D	10

14. Course Delivery System Determination

14.1 Delivery Model

The course delivery model was selected based on learner profile, program complexity, and learning objectives.

The program is delivered through:

- Instructor-led training (ILT) sessions (online/classroom)
- Interactive learning activities and case discussions
- Practical exercises using statistical tools
- Project-based learning approach (if applicable)

14.2 Rationale for Delivery Approach

The delivery approach was chosen because:

- Master Black Belt learners require interaction, coaching, and discussion-based learning.
- Advanced statistical concepts require guided facilitation and hands-on practice.
- Peer learning and group discussions enhance application and critical thinking.
- Real-world case studies support higher-level learning outcomes.

15. Learning Strategy and Methodology

The instructional methodology includes:

- Interactive lectures
- Guided discussions
- Case study analysis
- Group activities
- Simulation-based problem solving
- Hands-on statistical practice (Minitab / Excel / other tools)
- Formative quizzes
- Assignments and project-based evaluation

The instructional design incorporates adult learning principles and supports active learning and application-based skill building.

16.2 Pass Criteria

Candidates must achieve:

- Minimum final exam score: 70%

17. Final Exam Development Process

The final examination was developed using a structured and controlled process to ensure fairness, validity, and alignment with learning outcomes.

The exam development process includes:

- Development of an exam blueprint aligned with ILOs and module weightage
- SME drafting of exam questions based on approved content
- Review by an assessment panel to ensure clarity, correct answer validation, difficulty balance, and absence of bias
- Trial review and revisions
- Approval and secure storage of finalized question bank

18. Exam Blueprint

18.1 Blueprint Table

Module / Topic Area	Weightage %	No. of Questions
Module 1: Strategy & Deployment	10%	5
Module 2: Advanced DMAIC	36%	18
Module 3: Advanced Statistics	52%	26
Module 4: Lean Enterprise	0%	0
Module 5: Leadership & Mentoring	8%	4
Module 6: Sustainability & Governance	4%	2
Module 7: Case Studies	14%	7
TOTAL	100%	50 Questions

18.2 Bloom's Taxonomy Coverage

Cognitive Level	Coverage	No. of Questions (out of 50)
Knowledge & Understanding	22%	11
Application	30%	15
Analysis	28%	14
Evaluation	20%	10
TOTAL	100%	50 Questions

19. Item Writing and Item Banking Process

19.1 Item Writing Process

IMC develops exam questions based on program modules and Intended Learning Outcomes (ILOs). Questions are prepared by qualified SMEs/trainers using standard item-writing practices to ensure clarity and relevance. Each question is designed with one correct answer and appropriate distractors. Questions are written at different difficulty levels and are reviewed internally before final approval.

19.2 Item Banking Process

IMC maintains a controlled question bank for the MBB program. Questions are categorized by module and topic area. Access to the question bank is restricted to authorized personnel only. The question bank is periodically reviewed to ensure accuracy and alignment with current program content. Outdated questions are removed or archived to maintain exam integrity.

20. Program Implementation Process

The MBB program is implemented through a structured delivery process. This includes trainer onboarding, course schedule planning, learner registration, and distribution of learning materials such as slides, templates, and workbooks. Learners are provided with course orientation prior to delivery. Training is delivered as per the approved module plan, including activities and practice exercises. A final exam is conducted under controlled conditions, and certificates are issued only to learners who successfully meet completion and assessment requirements.

21. Program Evaluation and Continuous Improvement

The program is evaluated through a structured continuous improvement approach.

21.1 Evaluation Tools

Program effectiveness is measured using:

- Learner feedback surveys (end-of-program evaluation)
- Trainer evaluation reports (Conducted internally)
- Completion and pass rate analysis
- Corporate client feedback (for customized batches)
- Complaints and appeals register

21.2 Continuous Improvement Mechanism

Program improvements are implemented based on:

- Annual review meetings
- Learner performance data analysis
- Updates in Six Sigma best practices and industry trends
- Feedback from SMEs and trainers

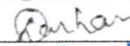
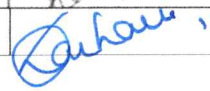
All changes are documented through version control and approval process.

22. Supporting Evidence Documents

The following documents support this IDP and are maintained as part of the accreditation evidence pack:

- Program brochure / course overview
- Curriculum outline and module structure
- Training schedule / session plan
- Trainer CVs and appointment letters
- Assessment blueprint document
- Sample exam paper and marking scheme
- Sample assignments and case studies
- Learner evaluation forms
- Program review report
- Certificate sample format

23. Approval and Sign-Off

Role	Name	Signature	Date
Prepared By	Bhuvaneshwari		3-Feb-2026
Reviewed By	Farhan Ahmad		3-Feb-2026
Approved By	Farhan Ahmad		6-Feb-2026
Approved By	Farhan Ahmad		13 - Apr-2026